EEG-Based Dementia Subtype Classification

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1. Problem Statement

Alzheimer's disease (AD) and frontotemporal dementia (FTD) are two of the most common forms of dementia. Differentiating between them is clinically important but challenging, especially in early stages where symptoms overlap. Current diagnosis relies on expensive neuroimaging and biomarkers that are not universally accessible (Alzheimer's Association, 2023).

Electroencephalography (EEG) is non-invasive, inexpensive, and captures neural activity patterns that differ between dementia subtypes. This project classifies resting-state EEG recordings into three groups (AD, FTD, healthy controls) using the OpenNeuro ds004504 dataset (Miltiadous et al., 2023) and compares multiple deep learning architectures, including a 1D-CNN, a 2D-CNN on spectrograms, and a bidirectional LSTM, to determine which signal representation and model best discriminates between these conditions.

- **Dataset:** OpenNeuro ds004504, CC0 license.
- **Format:** 19-channel EEG (10-20 system), 500 Hz, BIDS-standard .set files.
- **Subjects:** 88 total (36 AD, 23 FTD, 29 healthy controls).

The dataset is class-imbalanced, particularly FTD with only 23 subjects. Preprocessed recordings (bandpass filtered, artifact-rejected via ASR and ICA) serve as the starting point.

2. Problem Setup

Table 1 summarizes the input and output variables for this classification task.

Variable	Type	Shape	Encoding	Role
EEG Segments	Time series	19 ch × 2000	Z-score normalized	Input
Spectrograms	2D image	19 × freq × time	Log-scaled, [0,1]	Input (2D-CNN)
Diagnosis	Categorical	{AD, FTD, CN}	Integer (0, 1, 2)	Output

Table 1. Input and output variable summary.

Continuous EEG is segmented into fixed-length 4-second windows with 2-second overlap. Subject-level train/validation/test splits (61/13/14 subjects) prevent data leakage between sets. Macro F1 is the primary metric because accuracy is misleading for imbalanced classes; it would allow a model to score well by ignoring the minority FTD class entirely. ROC-AUC is reported for binary screening because it evaluates ranking quality independent of threshold choice.

3. Data Loading and Inspection

The dataset contains 88 subjects: 36 diagnosed with AD (mean MMSE 17.75, sd=4.5), 23 with FTD (mean MMSE 22.17, sd=8.22), and 29 healthy controls (MMSE 30). Age distributions overlap across groups (AD

mean 66.4, FTD 63.6, CN 67.9), reducing age as a confound. MMSE scores confirm expected cognitive separation, with AD showing the most severe impairment.

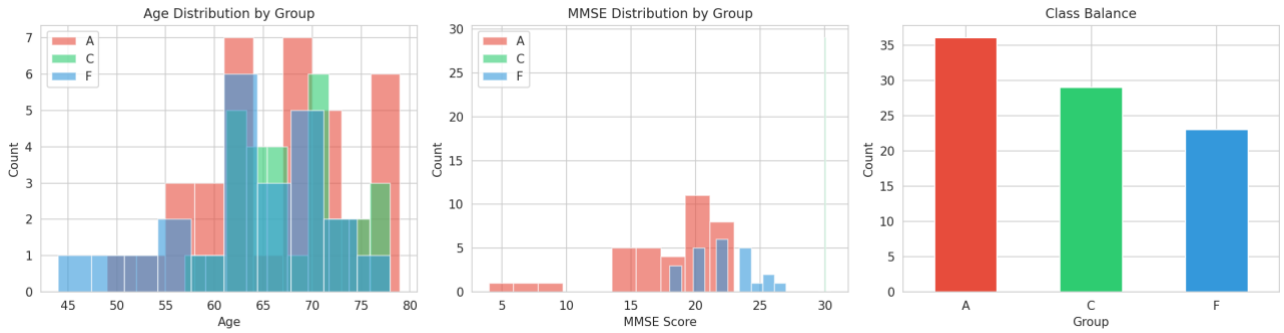

Figure 1. *Age distribution, MMSE distribution, and class balance across the three diagnostic groups.*

Recording durations varied: AD averaged 13.5 minutes, FTD 10.6 minutes, and CN 13.2 minutes. All recordings used 19 channels at 500 Hz. The shorter FTD recordings produce fewer segments per subject, which contributes to the class imbalance at the segment level.

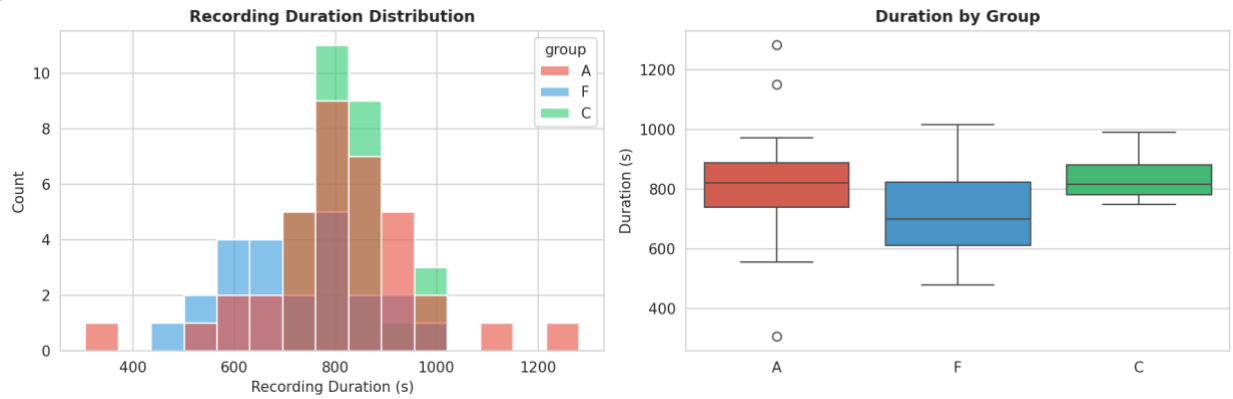

Figure 2. *Recording duration distributions by group.*

4. Data Exploration

The goal of the exploratory analysis was to identify spectral signatures that differentiate AD, FTD, and healthy controls before building classification models. AD is typically characterized by posterior alpha slowing and increased theta activity, while FTD often shows frontal theta/delta increases with relatively preserved posterior alpha.

4.1 Power Spectral Density

Power spectral density (PSD) analysis, computed via Welch's method, decomposed the EEG signal into canonical frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–45 Hz). Healthy controls exhibited a distinct posterior alpha peak (~10 Hz) that was diminished in both AD and FTD. Critically, the PSD distributions of AD and FTD were nearly identical across all bands, providing the first evidence that three-class separation may be fundamentally limited.

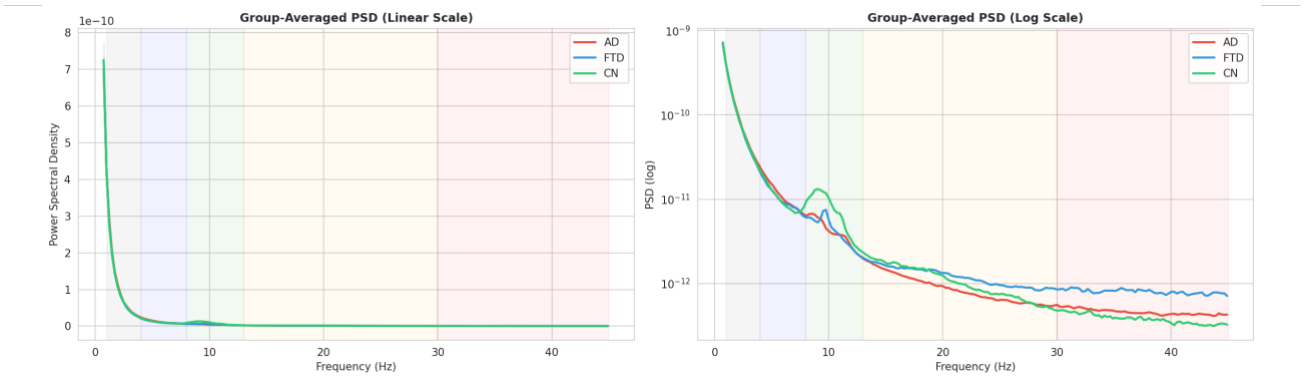

Figure 3. *Group-averaged power spectral density (linear and log scale) with frequency band shading. Note the diminished alpha peak in AD and FTD relative to CN, and the near-identical AD and FTD spectra.*

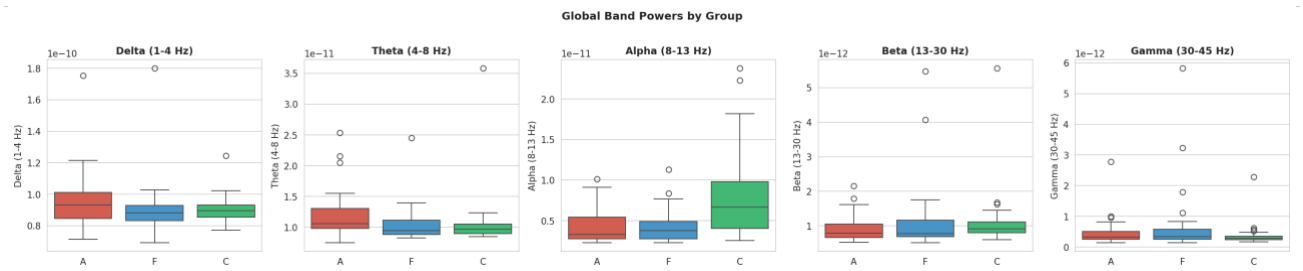

Figure 4. *Global band power boxplots by group across all five frequency bands.*

4.2 Regional Analysis

Channels were grouped into frontal (Fp1, Fp2, F3, F4, F7, F8, Fz), central (C3, C4, Cz), temporal (T3, T4, T5, T6), parietal (P3, P4, Pz), and occipital (O1, O2) regions. Occipital alpha showed the clearest group separation. Despite FTD being a frontal-lobe disease, frontal EEG features did not meaningfully separate FTD from AD in this resting-state data, possibly because resting-state recordings without task demands do not activate the frontal networks that FTD disrupts.

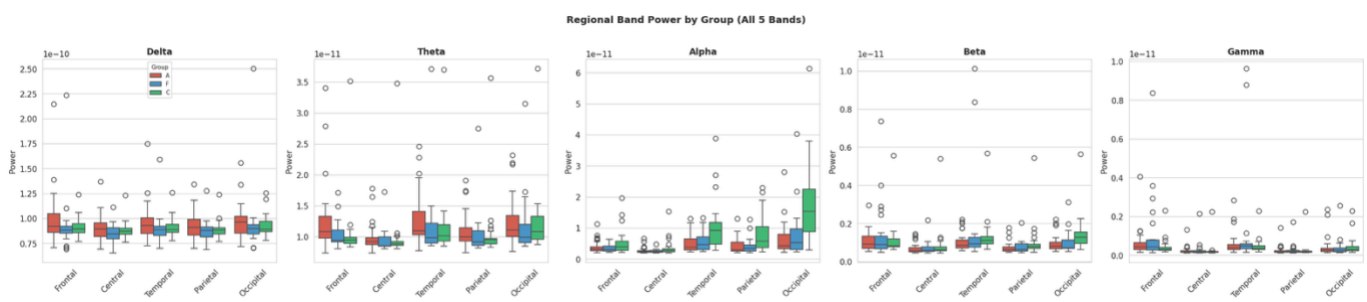

Figure 5. *Regional band power by group across all five frequency bands and five brain regions.*

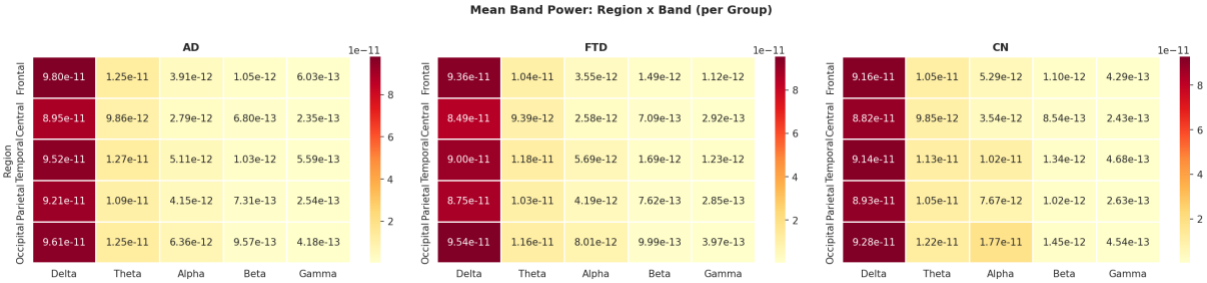

Figure 6. *Mean band power heatmaps (region × band) for AD, FTD, and CN groups.*

4.3 Ratio Analysis and Statistical Testing

Derived ratios were computed to target known AD vs. FTD biomarkers: theta/alpha ratio (TAR) per region, frontal-theta/temporal-alpha (FT_TA), temporo-parietal ATR, DTABR, and frontal beta/theta ratio. Kruskal-Wallis tests on the alpha/theta ratio showed significant overall group differences ($p < 0.0001$). Pairwise Mann-Whitney U tests found AD vs. CN and FTD vs. CN were both highly significant ($p < 0.001$), but AD vs. FTD was not statistically significant ($p = 0.13$). This result, established before any model was trained, predicted the eventual modeling outcome.

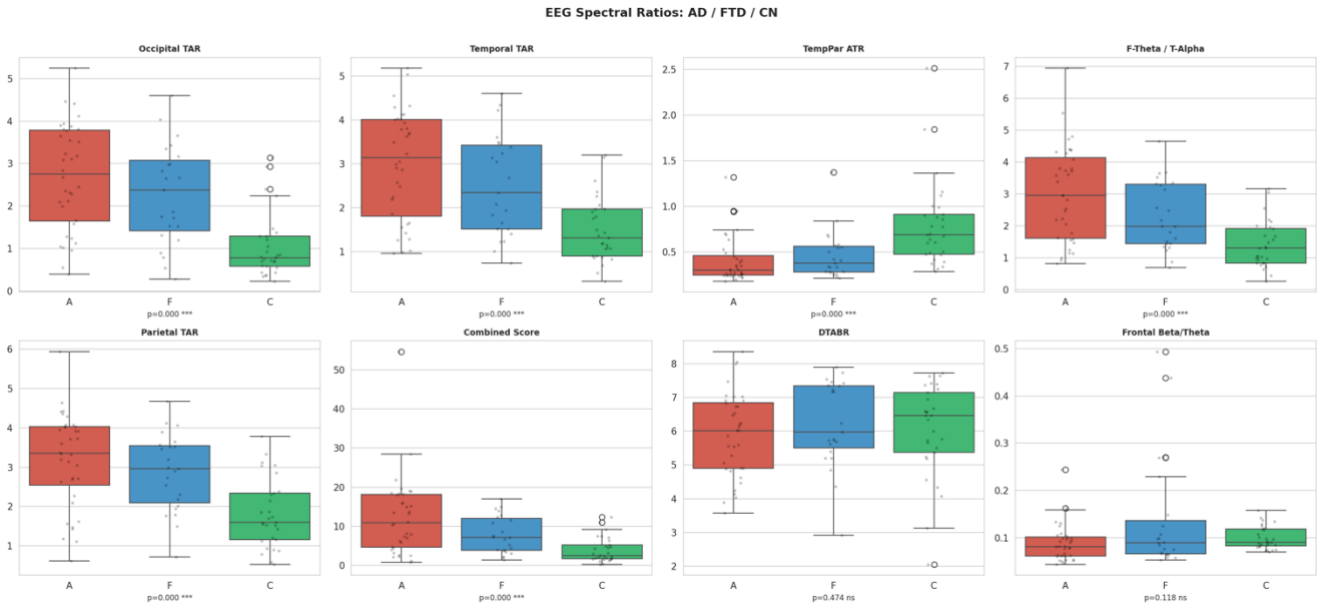

Figure 7. *Key spectral ratio boxplots with statistical significance annotations (2×4 grid). Includes TAR, FT_TA, DTABR, and combined scores.*

4.4 AD vs. FTD Separability

Cohen's d effect sizes between AD and FTD were uniformly small ($d < 0.35$) across all derived features. ROC analysis of individual features for AD-vs-FTD discrimination yielded AUCs near 0.5 (chance level), confirming that no single spectral feature reliably separates the two dementia subtypes. The MMSE correlation with alpha/theta ratio ($r = 0.51$, $p < 0.001$) further suggests cognitive decline manifests as a continuum in EEG rather than discrete diagnostic categories.

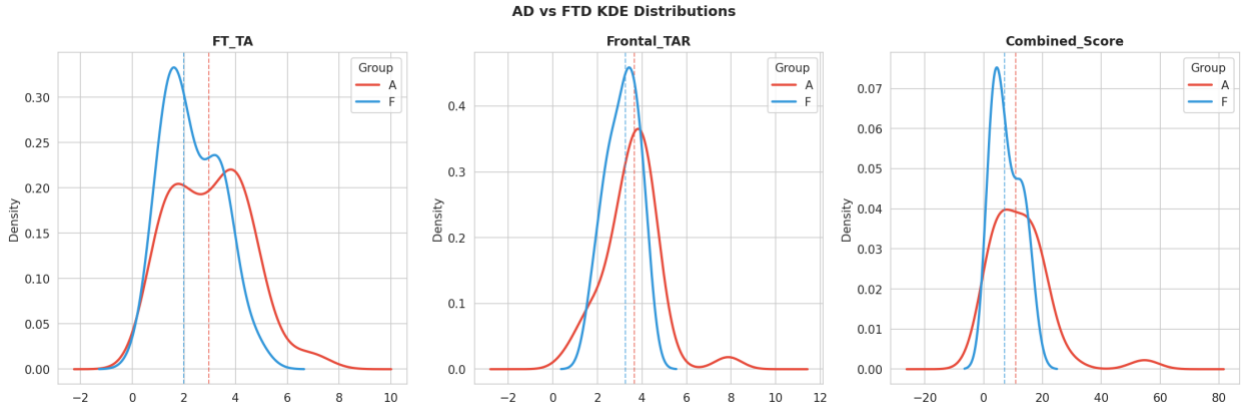

Figure 8. *AD vs. FTD KDE distribution for key ratio features (FT_TA, Frontal TAR, Combined Score) and.*

5. Data Preprocessing

Segmentation: Continuous EEG recordings (5–15 minutes each) were segmented into 4-second windows (2000 samples at 500 Hz) with 2-second overlap (50%). Window length balances temporal resolution with frequency content; at 4 seconds, the lowest resolvable frequency is 0.25 Hz, well below the delta band lower bound. The 50% overlap augments the dataset without excessive redundancy, yielding approximately 34,788 total segments.

Normalization: Z-score normalization was applied per-channel because EEG channels have different baseline amplitudes depending on cortical source and reference electrode distance. Statistics (mean, standard deviation) were computed from the training set only and applied identically to validation and test sets to prevent information leakage.

Subject-level split: Data was split at the subject level (61 train, 13 validation, 14 test) rather than the segment level. EEG segments from the same individual are highly correlated due to shared skull geometry, baseline rhythms, and noise characteristics. Segment-level splitting would allow the model to memorize individual patients and produce inflated performance estimates, a well-documented pitfall in EEG classification (Lee et al., 2023). The split was stratified by diagnosis to maintain class proportions.

Class imbalance: Addressed through weighted Focal Loss (γ=2.0) with class weights inversely proportional to class frequency (AD=0.773, FTD=1.284, CN=0.943). Focal Loss down-weights confident, easy predictions and focuses training on hard examples, which is particularly relevant for the underrepresented FTD class.

6. Baseline Models

Traditional ML models establish a performance floor against which deep learning is compared. For each 4-second segment, 296 features were extracted: per-channel statistics (mean, std, min, max), Hjorth parameters (activity, mobility, complexity), FFT band powers in five bands, spectral entropy, dominant frequency, per-channel alpha/theta ratio, and 11 cross-channel ratio features derived from the exploratory analysis.

Random Forest (200 trees, max depth 10, balanced class weights) achieved validation accuracy of 46.4% and macro F1 of 0.426. FTD recall was particularly poor at 15%. **Linear SVM** (balanced class weights) performed better at 55.6% accuracy and 0.539 F1 (ROC-AUC 0.713), with more balanced per-class predictions. Both models demonstrate that hand-crafted features capture some group differences but are insufficient for reliable classification, especially of FTD.

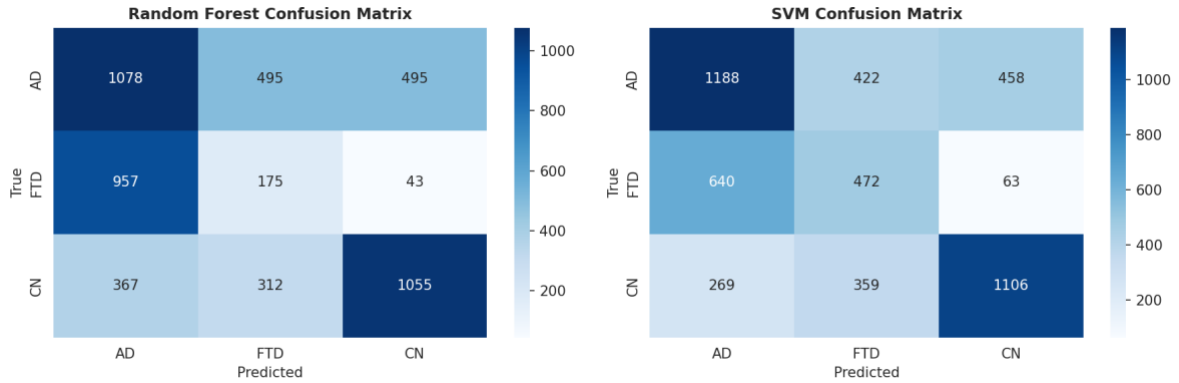

Figure 9. *Baseline model confusion matrices (Random Forest and SVM) on the validation set.*

7. Neural Network Implementation

Three architectures were compared, each representing a different hypothesis about which EEG patterns carry diagnostic information.

7.1 1D-CNN on Raw EEG

The 1D-CNN treats each EEG segment as a multivariate time series (19 input channels, 2000 time steps). The architecture uses a stem convolution (kernel size 25, stride 2) followed by two residual blocks with skip connections (64→128 channels, kernels 15 and 7), adaptive average pooling, and a fully connected classifier with dropout (0.5). The large initial kernel spans 50 ms at 500 Hz, capturing broad temporal context in EEG waveform morphology. Training used Adam (lr=1e-3), Focal Loss with class weights, CosineAnnealingLR (T_max=50, eta_min=1e-6), mild augmentation (Gaussian noise σ=0.05, amplitude scaling +/-10%), and early stopping on validation macro F1 (patience=10). The model achieved validation accuracy of 56.2% and macro F1 of 0.555 (AUC=0.786).

7.2 Bidirectional LSTM

A 2-layer bidirectional LSTM (hidden size 128) processed the EEG downsampled 4x to 500 timesteps. The forward and backward final hidden states were concatenated for classification. Validation F1 was 0.474, substantially below the CNN. This negative result is informative: resting-state EEG with eyes closed lacks the sequential event structure that LSTMs are designed to exploit. The diagnostic signal resides in local waveform morphology rather than long-range temporal dependencies.

7.3 2D-CNN on Spectrograms

STFT spectrograms were generated per channel (nperseg=128, noverlap=96, 1–45 Hz, log-scaled, normalized). A 3-block 2D-CNN (19-channel input, 32→64→128 filters) achieved validation F1 of 0.560 (AUC=0.735). While the 2D-CNN matched the 1D-CNN's topline F1, it exhibited systematic bias: 523 CN segments were misclassified as FTD, and its lower AUC indicated weaker overall ranking ability.

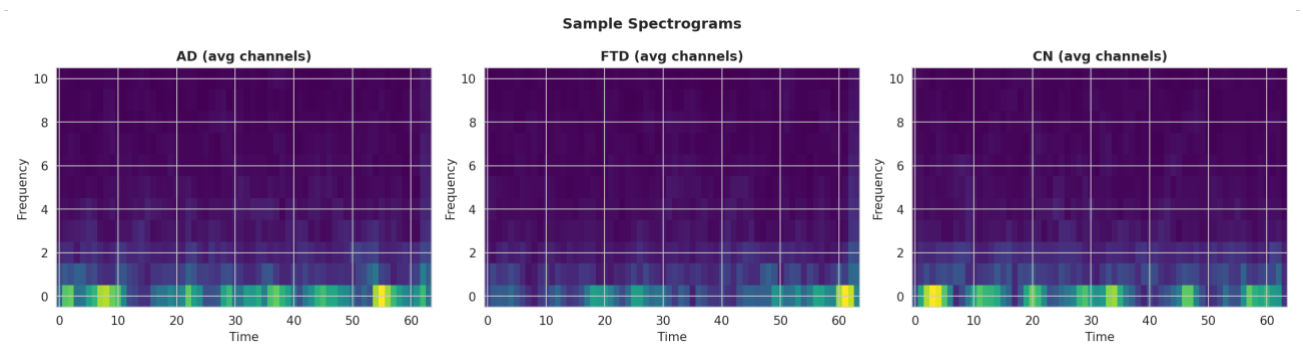

Figure 10. *Sample spectrograms for AD, FTD, and CN segments (channel-averaged).*

7.4 Model Comparison

The 1D-CNN was selected for refinement based on three factors: superior AUC (0.786 vs. 0.735), more balanced error distribution across classes, and a richer augmentation space for raw EEG. Its clear overfitting pattern (train accuracy ~93%, validation ~56%) provided a specific, addressable target. It is worth noting that in a pure clinical screening context, the 2D-CNN's error profile, predicting dementia over healthy, is the safer failure mode, as false negatives carry greater clinical risk than false positives. However, the 1D-CNN's

higher AUC indicates stronger underlying discriminative signal, making it the better candidate for iterative improvement given the constraints of this project.

Model	Val Accuracy	Val Macro F1	Val AUC	Parameters
Random Forest	0.464	0.426	N/A	296 features
Linear SVM	0.556	0.539	0.713	296 features
1D-CNN (selected)	0.562	0.555	0.786	~646K
LSTM	0.476	0.474	0.698	~500K
2D-CNN	0.561	0.560	0.735	~350K

Table 2. Performance comparison across all models on the validation set.

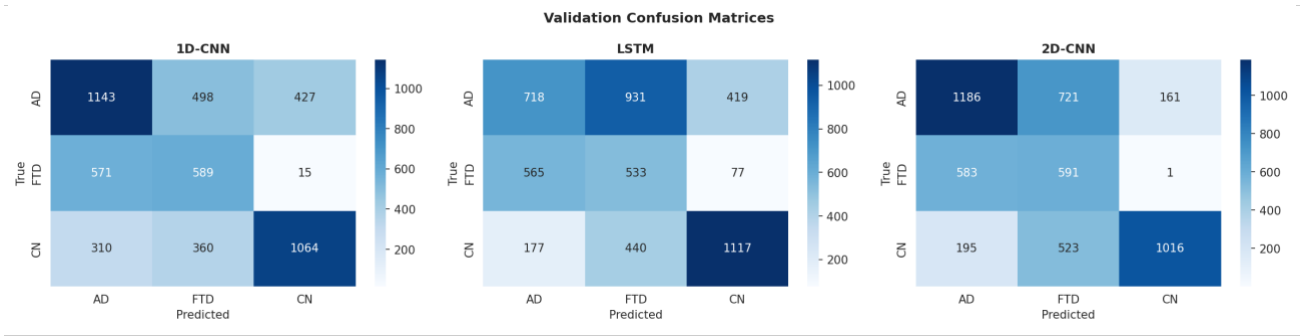

Figure 11. *Validation confusion matrices for all three deep learning models.*

8. Model Refinement

Refinement was guided by the continued overfitting. The 1D-CNN's learning curves showed a clear pattern: training loss continued decreasing while validation loss increased after epoch ~10. The root cause is ~646,000 parameters learning from only 61 unique training subjects, even though those subjects produce ~24,000 training segments.

8.1 Hyperparameter Search

A targeted 3-configuration grid varied weight decay (1e-4 to 1e-3) and toggled augmentation on/off. Removing augmentation improved validation F1 from 0.555 to 0.561, revealing that even mild noise injection and amplitude scaling were slightly harmful for this dataset. Increasing weight decay reduced the train-validation accuracy gap but did not raise the performance ceiling.

8.2 Mixup Regularization

Mixup (α=0.2) was applied on top of the best hyperparameter configuration. Unlike traditional augmentation, Mixup interpolates between pairs of training examples and their labels, creating virtual samples that smooth the decision boundary without altering raw EEG signals. This improved validation F1 to 0.576 and AUC to 0.790, the highest of any configuration tested.

Configuration	Val Accuracy	Val Macro F1	Val AUC
Original 1D-CNN	0.562	0.555	0.786
Best HP (no aug, wd=1e-4)	0.573	0.561	0.744
Mixup + Best HP (final)	0.589	0.576	0.790

Table 3. Refinement comparison on the validation set.

No refinement iteration dramatically exceeded the initial baseline, which is informative: the performance ceiling appears to be set by the data (88 subjects, overlapping AD/FTD signatures) rather than the model architecture or training procedure.

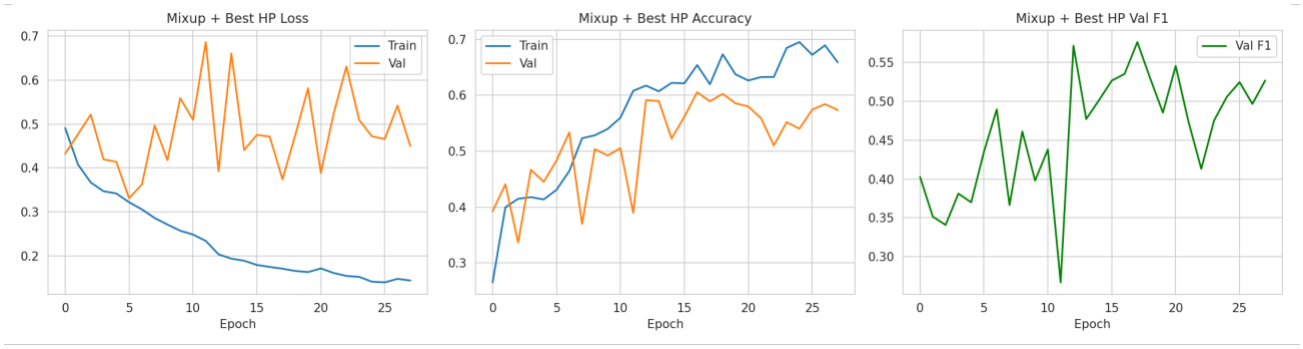

Figure 12. *Llearning curves showing overfitting reduction with Mixup.*

9. Final Evaluation

The final Mixup model was evaluated on the held-out test set (14 subjects, 5,662 segments) at two levels: segment-level (each 4-second window classified independently) and subject-level (softmax probabilities averaged across all segments per subject, with the highest mean probability class assigned).

Segment-level: accuracy 64.9%, macro F1 0.550, ROC-AUC 0.755. AD segments were classified with 73% recall and 79% precision. CN segments achieved 81% recall. FTD segments had only 15% recall, with most misclassified as AD.

Subject-level: 71.4% accuracy (10/14 correct). AD: 5/6 correct. CN: 5/5 correct. FTD: 0/3 correct. All three FTD subjects were misclassified (two as AD and one as CN), directly confirming the p=0.13 statistical finding. Literature results on this same dataset report approximately 80% for three-class classification (Khan et al., 2025); the present result of 71.4% is competitive given the 88-subject constraint.

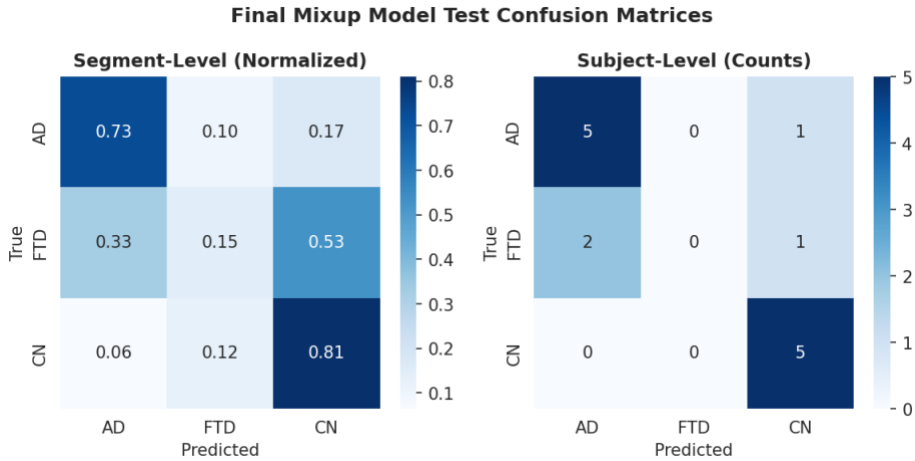

Figure 13. *Test set confusion matrices (segment-level normalized and subject-level counts) for the final Mixup model.*

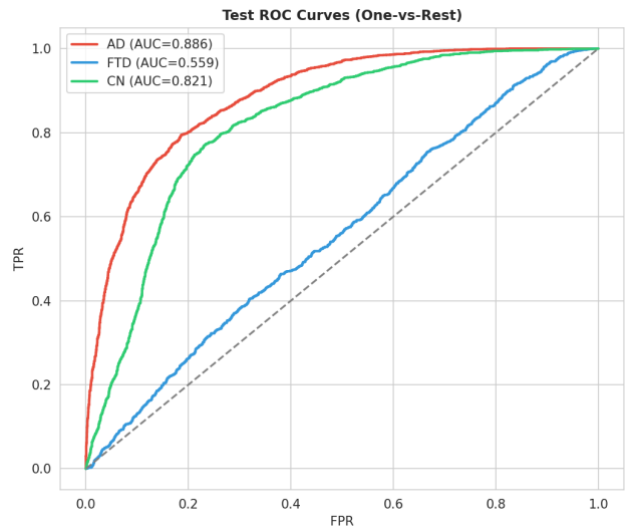

Figure 14. *Test set ROC curves (one-vs-rest) showing per-class AUC values.*

9.1 Per-Subject Error Analysis

Table 4 shows per-subject predictions sorted by model confidence. Correctly classified subjects had mean confidence of 0.611. Among the four misclassified subjects (3 FTD, 1 AD), sub-080 (FTD, predicted CN) had the highest confidence at 0.784 with 99.6% segment agreement, meaning the model was highly consistent in its wrong answer. Sub-077 (FTD, predicted AD) had the lowest confidence at 0.371, indicating genuine uncertainty. These patterns suggest some errors reflect true ambiguity in the EEG signal rather than model pathology.

Subject	True	Predicted	Correct	Confidence	Agreement	Segments	Note
sub-077	FTD	AD	No	0.371	0.437	348	Low confidence
sub-019	AD	AD	Yes	0.393	0.491	458	
sub-023	AD	CN	No	0.457	0.570	416	AD missed
sub-085	FTD	AD	No	0.547	0.720	279	
sub-028	AD	AD	Yes	0.574	0.767	408	
sub-026	AD	AD	Yes	0.647	0.935	448	
sub-029	AD	AD	Yes	0.729	0.984	367	
sub-080	FTD	CN	No	0.784	0.996	456	High confidence error
sub-054	CN	CN	Yes	0.792	0.988	417	

Table 4. Per-subject prediction analysis (sample of subjects, sorted by confidence).

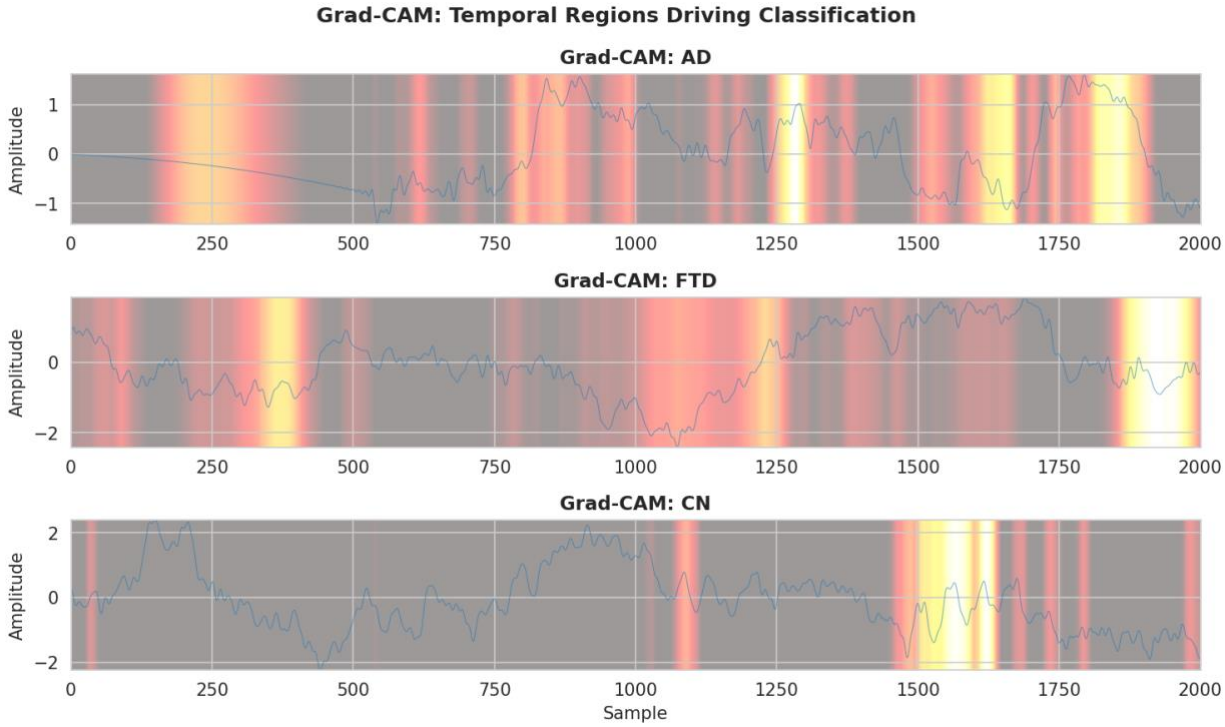

Figure 15. *Grad-CAM visualization showing temporal regions of the EEG signal most influential in the model's classification decisions for AD, FTD, and CN examples.*

11. Summary and Conclusions

Best model: 1D-CNN with Mixup regularization (α=0.2). Among three deep learning architectures, the 1D-CNN (F1=0.555, AUC=0.786) and 2D-CNN (F1=0.560, AUC=0.735) performed similarly at the segment level, while the LSTM lagged (F1=0.474). Refinement via hyperparameter search and Mixup improved the 1D-CNN to F1=0.576 and AUC=0.790. On the test set, the model achieved 64.9% segment accuracy and 71.4% subject-level accuracy (10/14).

FTD is the primary failure. All three FTD test subjects were misclassified. With only 23 FTD subjects total (3 in test) and a non-significant AD-vs-FTD statistical difference (p=0.13), the model lacks both the data and the signal to learn FTD-specific patterns.

Key insights: (1) Statistical exploration and sample size predicted the modeling outcome before training began. (2) The performance ceiling is data-limited: no architecture or hyperparameter change substantially exceeded the baseline. (3) Mixup regularization was more effective than traditional EEG augmentation (noise, scaling, channel dropout). (4) Calibration matters for screening: the model with the best average metrics was not the best screening tool due to overconfidence on a hard case.

Limitations: 88 subjects from a single hospital, no external validation, resting-state protocol only, and 14 test subjects means individual results carry high variance.

Future work: Pretrained EEG foundation models such as LaBraM (Jiang et al., 2024), which learned from 2,500 hours of EEG across 20 datasets, could provide feature representations that this small dataset cannot learn from scratch. The CAUEEG dataset (Kim et al., 2023) with 1,155 subjects spanning normal cognition, mild cognitive impairment, and dementia offers a direct scale-up opportunity. Functional connectivity features that capture network-level disruptions may better distinguish FTD from AD than the spectral power features used here.

References

Alzheimer's Association. (2023). 2023 Alzheimer's Disease Facts and Figures. Alzheimer's & Dementia, 19(4), 1598–1695.

Jiang, W.-B., Zhao, L.-M., & Lu, B.-L. (2024). Large Brain Model for Learning Generic Representations with Tremendous EEG Data in BCI. In Proceedings of ICLR 2024.

Khan, W., et al. (2025). An Explainable and Efficient Deep Learning Framework for EEG-Based Diagnosis of Alzheimer's Disease and Frontotemporal Dementia. Frontiers in Medicine.

Kim, M., Youn, Y. C., & Paik, J. (2023). Deep Learning-Based EEG Analysis to Classify Normal, Mild Cognitive Impairment, and Dementia: Algorithms and Dataset. NeuroImage, 272, 120054.

Lawhern, V. J., et al. (2018). EEGNet: A Compact Convolutional Neural Network for EEG-Based Brain-Computer Interfaces. Journal of Neural Engineering, 15(5), 056013.

Lee, H. T., et al. (2023). Risk of Data Leakage in Estimating the Diagnostic Performance of a Deep-Learning-Based Computer-Aided System for Psychiatric Disorders. Scientific Reports, 13, 16633.

Miltiadous, A., et al. (2023). A Dataset of Scalp EEG Recordings of Alzheimer's Disease, Frontotemporal Dementia and Healthy Subjects from Routine EEG. Data, 8(6), 95.